D Y PATIL DEEMED TO BE UNIVERSITY — RAMRAO ADIK — INSTITUTE OF TECHNOLOGY NAVI MUMBAI		**B.TECH IN ARTIFICIAL INTELLIGENCE (AI) AND DATA SCIENCE**						**SECOND YEAR SEM- III**		
Course Code	**Course Name**	**Teaching Scheme (Contact Hours 26)**						**Credits Assigned**		
		Theory				Practical		Theory		Total
2413ADU3T1	Embedded Systems	02				--		02		**02**
		Theory Evaluation Scheme (Marks)								
		Internal Assessment **(30)**						End Sem. Exam	End Sem. Exam Hrs.	Total
		Best 1 **(10)**		**(20)**						
		CA1	CA2	Assignment / Tutorial	Quiz / Seminar	Open Book Test / Surprise Test /Capstone Project				
		10	10	10	05	05		**60**	02	**90**

- **Prerequisite:**
  Basic knowledge of electronics and programming

- **Course Description:**
  This content covers digital systems and introduces embedded systems, emphasizing their distinctions from general-purpose computers, classifications, applications, and essential components such as microcontrollers, ASICs, RISC, and CISC architectures. It explores the 8051 microcontrollers, including its architecture, addressing modes, assembly and C programming, and peripheral interfacing with devices like LCDs and motors. Additionally, it provides insights into ARM processors and advanced ARM architectures.

- **Course Objectives:**
  - To establish a fundamental understanding of digital systems.
  - To provide an overview of embedded systems and embedded core concepts.
  - To introduce the fundamentals of the 8051 microcontroller and its assembly language programming.
  - To develop proficiency in assembly and high-level programming while exploring the integrated hardware of the 8051 microcontrollers.

- **Course Outcomes:** After completion of this course, learners will be able to:
  CO1: Understand number systems, logic gates, and design basic combinational and sequential circuits.
  CO2: Explain embedded systems, their classifications, applications, and key architectural concepts.
  CO3: Analyze 8051 architecture and apply assembly programming concepts.
  CO4: Develop programs using I/O ports, timers, serial communication, and interrupts.
  CO5: Design and implement 8051-based peripheral interfacing and real-world embedded applications.
  CO6: Explain ARM architecture, register operations, pipelining, and evaluate Cortex processor architecture.

Module No.	Detailed Content	Hrs (26)	CO
1	**Overview of Digital Systems** Number Systems: Introduction to Binary, Octal, Decimal, and Hexadecimal number systems, conversion methods. **Logic Gates**: Representation of basic logic gates with symbols and truth tables. Circuit diagrams and working principles of Adders and Multiplexers, De-multiplexers, R-S Flip-Flops, and D Flip-Flops.	5	CO1
2	**Introduction to Embedded systems** Embedded Systems and general-purpose computer systems, Comparison of microprocessor and microcontroller, classifications, applications. ASIC, RISC and CISC architecture.	4	CO2
3	**MCS-51 microcontroller** Family and Architecture of 8051, Introduction to Assembly language programming, call, loop, jump instructions.	4	CO3
4	**8051 Addressing Modes & Programming** I/O port programming, Addressing modes, Arithmetic and Logic Instructions and programs. Timer/ counter, Concepts of Serial Communication, RS232, 8051 interrupts.	5	CO4
5	**8051 Peripheral Interfacing** Interfacing 8051 with peripherals such as a keyboard, seven-segment display, LCD, DC motor, and stepper motor. Case studies on embedded systems: digital cameras and washing machines.	4	CO5
6	**Introduction to ARM** ARM Architecture: Overview of architecture, register set, operating modes, and CPSR, pipelining in ARM processors. **Advanced ARM Processors:** Introduction to Cortex processor core and its architecture.	4	CO6

Text books:

1. Digital Design: With an Introduction to the Verilog HDL, VHDL, and System Verilog, 6e, 2018.
2. Digital Fundamentals, 11th Edition by Thomas L. Floyd, Pearson, 2017.
3. The 8051 Microcontroller and Embedded systems by Mazidi M.A, Pearson Education, Second edition, 2007.
4. The 8051 Microcontroller by Kenneth Ayala, Thomson Delmar Learning, Third Edition, 2007.
5. Embedded Systems: Architecture, Programming and Design by Rajkamal, McGraw Hill Education (India) Private Limited, New Delhi, Third Edition, 2017.
6. Embedded Systems Design by Steve Heath, Newness publication, Second edition, 2003.
7. Embedded Software Primer by David Simon, Pearson Education, 2002.
8. Cortex-M4 Devices Generic User Guide, 2011.
9. Embedded System: Real-Time Interfacing to ARM Cortex-M Microcontrollers by Jonathan W. Valvano, Volume-2, Fourth edition, 2014.
10. ARM System Developers guide, Andrew N SLOSS, Dominic SYMES, Chris WRIGHT, Elsevier, 2012